Effective DPS cross section

$$\sigma_{eff} = \frac{1}{2}\frac{\sigma^2(J/\psi)}{\sigma_{DPS}^{\text{no mass cut}}}$$

$$X = B\,\sigma(J/\psi) = 10.63 \text{ nb}$$

$\varepsilon_m = 0.987339$ from DPS GEN-only event mixing

B cancels between X^2 and the pair-visible denominator

$$\sigma_{eff} = 2.48 \pm 0.86\,(\text{stat.})^{+0.25}_{-0.22}\,(\text{syst.})\ \text{mb}$$

Prompt J/psi: 10 < pT < 40 GeV, |y| < 2.0

Preliminary result

